



SAM PETER

Electronics and Communication Engineering Student

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Career Objective

Motivated Electronics and Communication Engineering student interested in AI/ML, Application Development, Embedded Systems, and VLSI Design. Passionate about building intelligent software solutions and solving real-world engineering problems through innovation and technology.

Education

NSS College of Engineering, Palakkad	2023 – 2027
Bachelor of Technology in Electronics and Communication Engineering	CGPA: 6.84
Kendriya Vidyalaya (CBSE)	2012 - 2023

Internship Experience

AI & Machine Learning Intern – Novola.ai & AccelerateX	June 2025
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- Built AI-based projects including Sentiment Analyzer and Hand Gesture Recognition System.
- Worked on ML models such as Linear Regression, Logistic Regression, Decision Trees, and Random Forest.
- Explored computer vision, feature optimization, model training, and dataset preprocessing.
- Gained hands-on experience in AI implementation methodologies and project development.

Academic Projects

8-bit Microprocessor Design	2026
<i>Verilog HDL, Xilinx Vivado</i>	

- Designed and implemented an 8-bit microprocessor with modular datapath architecture.
- Developed ALU, Control Unit, Register File, and instruction execution cycle.
- Integrated 24+ arithmetic and logical operations with status flag support.
- Verified processor functionality using Vivado simulations and waveform analysis.

AI Fish Tank Simulator	2026
<i>Python, OpenCV, MediaPipe, Pygame</i>	

- Built an AI-powered virtual aquarium with real-time gesture interaction.
- Implemented hand tracking using OpenCV and MediaPipe.
- Simulated fish behaviors including seek, flee, wander, and idle movement.
- Integrated vector physics and audio feedback using NumPy and gTTS.

Sentiment Analysis ML Web Application	2025
<i>Python, Machine Learning, Random Forest</i>	

- Built a sentiment classification web application using Random Forest.
- Trained on 1.6M+ samples achieving nearly 92% accuracy.
- Implemented preprocessing, feature tuning, and real-time prediction system.

Smart Automated Delivery Box System	2025
<i>Arduino Uno, ESP32, IoT</i>	

- Developed an IoT-enabled automated parcel management system.

- Implemented OTP verification, UART communication, and Telegram notifications.
- Added image verification and load-cell based parcel validation.
- Selected for YIP District Level 8.0 and Cognizant Technoverse Hackathon.

Water Taranga – IoT Monitoring System

2025

Python, Flask/Django, IoT

- Developed an IoT-based system for monitoring water and electricity usage.
- Implemented threshold-based alerts for excessive consumption detection.
- Built secure authentication using bcrypt and database integration.
- Designed responsive dashboard using HTML, CSS, JavaScript, and Python.

Certifications

- The Joy of Computing using Python – IIT Madras (NPTEL) – 2025
- Deep Learning – IIT Ropar (NPTEL) – 2026

Positions of Responsibility

Secretary, IEEE PES SBC NSSCE

2026 – Present

- Organized Nexotech IEEE Conference 2026 and Luminous Logic workshop.

Operations Lead, iEDC NSSCE

2026 – Present

- Coordinated District Level YIP 7.0 and YIP 8.0 and Kerala Innovation Festival 2025.

Technical Skills

Languages: Python, C, Embedded C, HTML, CSS

Technologies: AI/ML, OpenCV, MediaPipe, Flask, Arduino, ESP32

Tools: Xilinx Vivado, VS Code, Arduino IDE, Jupyter Notebook

Domains: Application Development, IoT, Computer Vision, VLSI

Technical Activities

- Participated in Bharatiya Antariksh Hackathon 2025.
- Developing Persistent Scene Change Detection Pipeline for Vehant Hackathon 2026.
- Presented Smart Automated Delivery Box System at YIP District Evaluation.

Additional Information

Languages: English, Hindi, Malayalam

Interests: Movies, Football, Cricket, Travelling